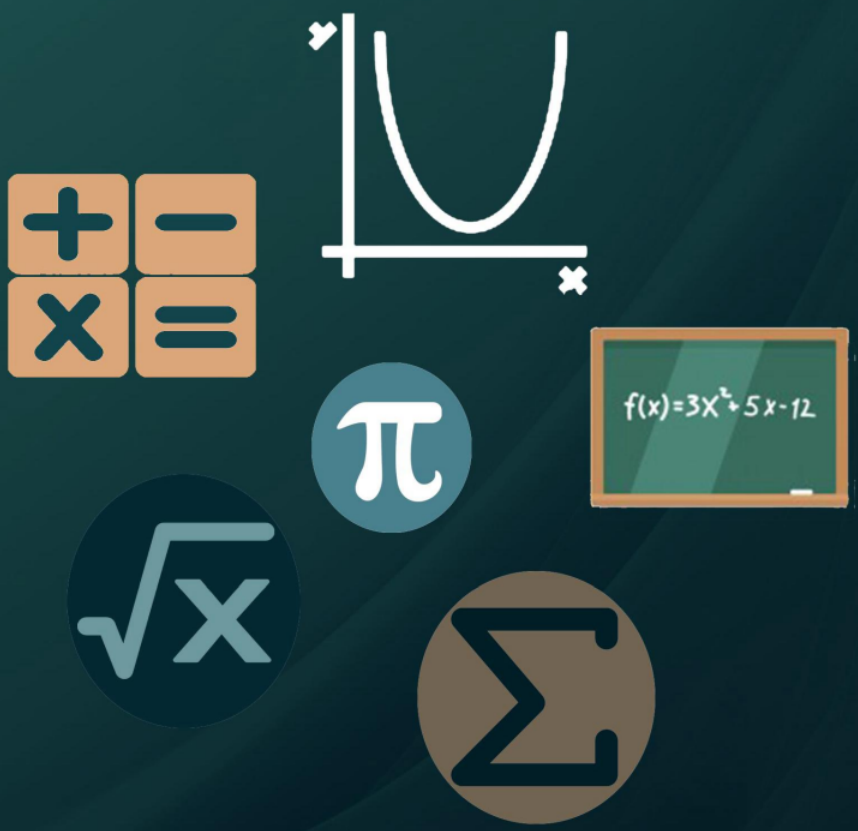
Sheet

MATHEMATICS

Sequence & Series



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Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 4
5.	Exercise 2	Page - 6
6.	Exercise 2A	Page - 10
7.	Answer Key	Page - 14

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction				
2	General Term of AP	1 - 4	1, 17	1, 2	
3	Summation of AP	5 - 9		3 - 9	1, 20, 21
4	Properties of AP	10 - 13	2, 18, 19	10 - 12	2
5	Arithmetic Mean	14 - 17	3	13 - 15	
6	General Term of GP	18 - 21	4		
7	Summation of GP	22 - 26	10 - 14	16 - 22	3, 4, 15, 22, 23
8	Properties of GP	27			5
9	Geometric Mean	28, 29		23	24, 25
10	General Term of HP			24	6, 16, 18
11	Harmonic Mean	30 - 32			7, 8
12	Relation Between AM, GM, HM	33 - 36	5, 7 - 9, 15, 20	25-30	9, 19, 26
13	Arithmetico Geometric Progression	37, 38	6, 21	31, 32	17, 27, 28
14	Special Sequence Type - 1	39		34	10, 12
15	Special Sequence Type - 2	40, 41	16, 22 - 24	33	11
16	Special Sequence Type - 3			35	16, 29
17	Special Sequence Type - 4				13, 14

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

General Term of Arithmetic Progression

Q1. If 6th and 11th terms of an AP are 17 and 32, then its 20th term is :-

(A) 78 (B) 79

(C) 59 (D) 60

Q2. The sum of the first and third term of an arithmetic progression is 12 and the product of first and second term is 24, then first term is :-

(A) 1 (B) 8

(C) 4 (D) 6

Q3. If the pth term of an AP is q and its qth term is p, then its (p+q)th term is :-

(A) – p (B) –q

(C) –(p + q) (D) 0

Q4. 15th term from the end in 4, 7, 10, 13,.....,154 is :-

(A) 46 (B) 112

(C) 321 (D) 199

Summation of Arithmetic Progression

Q5. In an A.P., first term is 1, last term is 11 and sum of n terms is 36, then n =

(A) 5 (B) 6

(C) 7 (D) none

Q6. If sum of n terms of an A.P. is $3n^2 + 5n$ & $T_m = 164$, then m equals :-

(A) 26 (B) 27

(C) 28 (D) 24

Q7. If m^{th} term of an A.P. is $\frac{1}{n}$ and n^{th} term is $\frac{1}{m}$ then find the sum of mn terms.

(A) $\frac{1}{2}(mn+1)$ (B) $\frac{1}{2}(mn-1)$

(C) mn/2 (D) mn

Q8. If nth term of an AP is $\frac{1}{3}(2n+1)$, then the sum of its first 19 terms is :-

(A) 131 (B) 132

(C) 133 (D) 134

Q9. The sum of all integers between 81 and 719 which are divisible by 5 is

(A) 51800 (B) 50800

(C) 52800 (D) None of these

Properties of Arithmetic Progression

Q10. Three numbers are in A.P. whose sum is 33 and product is 792, then the smallest number from these numbers is-

(A) 4 (B) 8

(C) 11 (D) 14

Q11. If a, b, c, d, e are in A.P. then the value of a - 4b + 6c - 4d + e is-

(A) 1 (B) 2

(C) 0 (D) None of these

Q12. If a, b, c are in AP, then $(a - c)^2$ equals :-

(A) $4(b^2 - ac)$ (B) $4(b^2 + ac)$

(C) $4b^2 - ac$ (D) $b^2 - 4ac$

Q13. If angles of quadrilateral are in A.P. with common difference 10° then smallest angle is :-

(A) 70° (B) 75°

(C) 80° (D) 85°

Arithmetic Mean

Q14. If the sum of the resulting series obtained by introducing n A.M.'s between 2 and 38 is 200, then n is equal to :-

(A) 10 (B) 9

(C) 8 (D) 6

Q15. n AM's are inserted between 7 and 71. If fifth AM is 27 then n is equal to :-

(A) 17 (B) 16

(C) 15 (D) 14

Q16. If 8 arithmetic means are placed between 2 and 17 then the 5th arithmetic mean will be:

(A) $\frac{61}{3}$ (B) $\frac{76}{3}$

(C) $\frac{31}{3}$ (D) $\frac{106}{3}$

Q17. Sum of 20 arithmetic means between 1 and 99 is:-

(A) 1500 (B) 1000

(C) 1150 (D) 2100

Geometric Progression

Q18. If x, 2x + 2 and 3x + 3 are in GP, then its 4th term is-

(A) 27 (B) 13.5

(C) – 27 (D) – 13.5

Q19. The sixth term of a G.P is 32 & its 8th term is 128, then its common ratio is :-

(A) –1 (B) 2

(C) 4 (D) –4

Q20. If $y = 1 + x + x^2 + ...$ up to infinite terms, then x is equal to

(A) $\frac{y-1}{y}$ (B) $\frac{1-y}{y}$

(C) $\frac{y}{1-y}$ (D) None

Q21. If n^{th} term of $5, \sqrt{5}, 1, ...$ is $\frac{1}{625}$, then the value of n is :-

(A) 7 (B) 10

(C) 11 (D) None of these

Summation of Geometric Progression

Q22. If first term of a G.P. is 1 and the sum of its 3rd and 5th term is 90 then the common ratio is-

(A) ± 4 (B) ± 3

(C) 2 (D) –2

Q23. Sum of 9 terms of G.P. $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + ...$ is-

(A) $\frac{171}{256}$ (B) $\frac{256}{171}$

(C) 171 (D) 256

Q24. The value of $5^{1/2} \cdot 5^{1/4} \cdot 5^{1/8} ..... \infty$ is -

(A) 5 (B) 25

(C) 1/5 (D) 1/25

Q25. If second term of a G.P. is 2 and the sum of its infinite terms is 8, then its first term is :-

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) 2 (D) 4

Q26. The first term of an infinite G.P. is 1 and every term is equals to the sum of the successive terms, then its fourth term will be:

(A) $\frac{1}{2}$ (B) $\frac{1}{8}$

(C) $\frac{1}{4}$ (D) $\frac{1}{16}$

Properties of Geometric Progression

Q27. If $(m+n)^{th}$ term of a G.P. is 9 and $(m-n)^{th}$ term is 4, then m^{th} term will be-

(A) 6 (B) $\frac{1}{6}$

(C) 6.5 (D) None

Geometric Mean

Q28. If 5 GM's be introduced between 486 and $\frac{2}{3}$, then 4th GM is

(A) 6 (B) –6

(C) 12 (D) –12

Q29. Find the product of three GM's between 4 & $\frac{1}{4}$:-

(A) 1 (B) 4

(C) $\frac{1}{4}$ (D) None of these

Harmonic Progression

Q30. If a, b, c are in HP, then $\frac{a-b}{b-c}$ is equal to

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$

(C) $\frac{a}{c}$ (D) $\frac{c}{b}$

Q31. If the 10th term of an HP is 21 and 21st term of the same HP is 10, then find the 210th term.

(A) 210 (B) 1/210

(C) 1 (D) 2

Q32. The 6th term of the series 2, $1\frac{3}{4}, 1\frac{5}{9}$, ... is :-

(A) 7/6 (B) 6/7

(C) 6/19 (D) 3/7

AM, GM, HM

Q33. For any numbers if A.M. = 16, H.M. = $\frac{63}{4}$ then G.M. will be-

(A) $6\sqrt{7}$ (B) $\sqrt{7}$

(C) $6\sqrt{3}$ (D) None of these

Q34. If a, b and c are positive real numbers then $\frac{a}{b}+\frac{b}{c}+\frac{c}{a}$ is greater than or equal to :-

(A) 3 (B) 6

(C) 27 (D) 5

Q35. If A_1, A_2, G_1, G_2 and H_1, H_2 are two A.M's, G.M's and H.M.'s between two numbers, then $\frac{A_1+A_2}{H_1+H_2}\cdot\frac{H_1H_2}{G_1G_2}$ equals to-

(A) 0 (B) 4

(C) 1 (D) 8

Q36. If a, b, c $\in R^+$ and a + b + c = 5 then maximum value of ab^3c is equal to :-

(A) 3 (B) 9

(C) 27 (D) 81

Arithmetic Geometric Progression (AGP)

Q37. The sum of the infinite terms of $1+\frac{4}{5}+\frac{7}{5^2}+\frac{10}{5^3}+.....$ will be-

(A) $\frac{35}{16}$ (B) $\frac{16}{35}$

(C) $\frac{15}{16}$ (D) $\frac{7}{4}$

Q38. If $|x| < 1$, then the sum of the series $1 + 2x + 3x^2 + 4x^3 + ... \infty$ will be-

(A) $\frac{1}{1-x}$ (B) $\frac{1}{1+x}$

(C) $\frac{1}{(1+x)^2}$ (D) $\frac{1}{(1-x)^2}$

Special Sequence Type - 1

Q39. The sum of n terms of the series whose n^{th} term is n(n + 1) is equal to-

(A) $\frac{n(n+1)(n+2)}{3}$

(B) $\frac{(n+1)(n+2)(n+3)}{12}$

(C) $n^2 (n + 2)$

(D) $n(n + 1)(n + 2)$

Special Sequence Type - 2

Q40. $2 + 4 + 7 + 11 + 16 + \ldots\ldots\ldots +$ up to n terms is equal to

(A) $\frac{1}{6}(n^2 + 3n + 8)$ (B) $\frac{n}{6}(n^2 + 3n + 8)$

(C) $\frac{1}{6}(n^2 - 3n + 8)$ (D) $\frac{n}{6}(n^2 - 3n + 8)$

Q41. $1 + 3 + 7 + 15 + 31 + \ldots\ldots +$ up to n terms is equal to

(A) $2^{n+1} - n$ (B) $2^{n+1} - n - 2$

(C) $2^n - n - 2$ (D) none of these

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The nth term of an AP is P^2 and the sum of the first n term is S^2. The first term is :-

(A) $\frac{2S^2 - P^2 n}{2n}$

(B) If P = 2, S = 10, n = 10, then first term is 16.

(C) $\frac{2S^2 - P^2 n}{n}$

(D) If P = 2, S = 10, n = 10, then first term is 8.

Q2. For the A.P. given by $a_1, a_2, \ldots\ldots\ldots, a_n, \ldots\ldots\ldots$, with non-zero common difference, the equations satisfied are :-

(A) $a_1 + 2a_2 + a_3 = 0$

(B) $a_1 - 2a_2 + a_3 = 0$

(C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$

(D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$

Q3. There are n AM's between 1 & 31 such that 7th mean : (n – 1)th mean = 5 : 9, then what is the value of n :-

(A) 14 (B) 6 (C) 12 (D) 13

Q4. If the pth term of a GP is x and q^{th} term is y then the nth term is :-

(A) $\left(\frac{x^{n-q}}{y^{n-p}}\right)^{\frac{1}{p-q}}$ (B) $\left(\frac{x^{n-p}}{y^{n-p}}\right)^{\frac{1}{p-q}}$

(C) $x\left(\frac{x}{y}\right)^{\frac{n-p}{p-q}}$ (D) $x\left(\frac{x}{y}\right)^{\frac{n-q}{p-q}}$

Q5. If a, b, c > 0 then $(a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$ can be :-

(A) 3 (B) 9

(C) 99 (D) 39

Q6. $\sum_{r=1}^{\infty}(2r-1)\left(\frac{9}{11}\right)^r$ is equal to :-

(A) 45

(B) 55

(C) sum of first nine natural numbers

(D) sum of first ten natural numbers

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 7 to 9)

Let $A_1, A_2, A_3, \ldots\ldots, A_m$ be arithmetic means between –2 and 1027 and $G_1, G_2, G_3, \ldots\ldots, G_n$ be geometric means between 1 and 1024. Product of geometric means is 2^{45} and sum of arithmetic means is 1025×171.

Q7. The value of n is :-
(A) 7 (B) 9
(C) 11 (D) None of these

Q8. The value of m is :-
(A) 340 (B) 342
(C) 344 (D) 346

Q9. The value of $G_1 + G_2 + G_3 + \ldots\ldots\ldots + G_n$ is :-
(A) 1022 (B) 2044
(C) 512 (D) None of these

Paragraph # 2 (Ques. 10 to 14)

There are $4n + 1$ terms in a sequence of which first $2n + 1$ are in Arithmetic Progression and last $2n + 1$ are in Geometric Progression the common difference of Arithmetic Progression is 2 and common ratio of Geometric Progression is 1/2. The middle term of the Arithmetic Progression is equal to middle term of Geometric Progression. Let middle term of the sequence is T_m and T_m is the sum of infinite Geometric Progression whose sum of first two terms is $\left(\frac{5}{4}\right)^2$ n and ratio of these terms is $\frac{9}{16}$.

On the basis of above information, answer the following questions :

Q10. Number of terms in the given sequence is equal to -
(A) 9 (B) 17 (C) 13 (D) none

Q11. Middle term of the given sequence, i.e. T_m is equal to -
(A) 16/7 (B) 32/7 (C) 48/7 (D) 16/9

Q12. First term of given sequence is equal to -
(A) –8/7, –20/7 (B) –36/7
(C) 36/7 (D) 48/7

Q13. Middle term of given A. P. is equal to -
(A) 6/7 (B) 10/7 (C) 78/7 (D) 11

Q14. Sum of the terms of given A. P. is equal to -
(A) 6/7 (B) 7 (C) 3 (D) 6

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15. The expression $\frac{x^{a+2} + y^{a+2}}{x^{a+1} + y^{a+1}}$ is :- $(x \neq y)$

Match the items of column I with that of column II.

Column I	**Column II**
(A) AM of x and y	(P) when $a = -2$
(B) GM of x and y	(Q) when $a = -\frac{3}{2}$
(C) HM of x and y	(R) when $a = -1$

Q.16 **Column-I** **Column-II**

(A) n^{th} term of the series 4, 11, 22, 37, 56, 79,.....is equal to- (P) $2n^2 + n$

(B) $|1^2 - 2^2 + 3^2 - 4^2 ..... 2n|$ terms is equal to- (Q) $2n^2 + n + 1$

(C) sum to n terms of the series 3, 7, 11, 15,..... is - (R) $-(n^2 + n)$

(S) $\frac{1}{2}(n^2 + n)$

SUBJECTIVE TYPE QUESTIONS

Q17. If p^{th}, q^{th} and r^{th} term of an A.P. are respectively a, b, and c then find the value of

$a(q - r) + b(r - p) + c(p - q)$.

Q18. If a^2, b^2, c^2 are in A.P. then prove that :-

(a) $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P.

(b) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in A.P.

Q19. If $\log_3 2, \log_3(2^x - 5)$ & $\log_3\left(2^x - \frac{7}{2}\right)$ are in AP, determine x.

Q20. Prove that a Δ ABC is equilateral if and only if $\tan A + \tan B + \tan C = 3\sqrt{3}$.

Q21. If the sum to infinity of the series $3 + (3 + d)\frac{1}{4} + (3 + 2d)\frac{1}{4^2} + ... \infty$ is $\frac{44}{9}$, then find d.

Q22. Compute the sum $(31)^2 + (32)^2 + (33)^2 + ... + (50)^2$.

Q23. If $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$ and p, q, r are in A.P. then prove that x, y, z are in H.P.

Q24. If $\sum_{r=1}^{n} I(r) = n(2n^2 + 9n + 13)$, then find the sum $\sum_{r=1}^{n} \sqrt{I(r)}$.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The number of terms common to the A.P.'s 3,7, 11..... 407 and 2, 9, 16, ... 709 is :-

(A) 51 (B) 14

(C) 21 (D) 28

Q2. If $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are the pth, q^{th}, r^{th} terms respectively of an A.P. then $ab(p-q) + bc(q-r) + ca(r-p)$ equals to-

(A) 1 (B) –1

(C) 0 (D) None of these

Summation of AP

Q3. The sum of n terms of two series in A.P. are in the ratio (3n – 13) : (5n + 21). Find the ratio of their 24th terms.

(A) 1 : 2 (B) 2 : 1

(C) 1 : 13 (D) 13 : 1

Q.4 The maximum sum of the series

$20+19\frac{1}{3}+18\frac{2}{3}+18+.....$ is :-

(A) 310 (B) 290

(C) 320 (D) None of these

Q5. If a is the first term , 1 is the common difference and b is the last term of an AP, then its sum is :-

(A) $\frac{(a+b)(1+a-b)}{2}$

(B) $\frac{(a+b)(1-a+b)}{2}$

(C) $\frac{(a+b)(1-a)}{2}$

(D) $(a+b)(1-a+b)$

Q6. In a polygon, interior angles are in A.P. with common difference 5° and smallest angle 120°, then the number of sides is :-

(A) 9 or 16 (B) 16

(C) 9 (D) None

Q7. If a_1, a_2, a_3 ..... a_{100} are in A.P. and $a_1 + a_2 + a_3 +$ ..... $a_{50} = \alpha$ and $a_{51} + a_{52} + a_{53} +$ ..... $a_{100} = \beta$ then common difference of A.P. is equal to :-

(A) $\beta - \alpha$ (B) $\frac{\beta-\alpha}{5000}$

(C) $\frac{\beta-\alpha}{2500}$ (D) $\frac{\beta-\alpha}{50}$

Q8. Let the sequence $a_1, a_2, a_3, ... , a_n, ...$ form an A.P. Then $a_1^2 - a_2^2 + a_3^2 - a_4^2 + ... + a_{2n-1}^2 - a_{2n}^2$ is equal to-

(A) $\frac{n}{2n-1}(a_1^2 - a_{2n}^2)$ (B) $\frac{2n}{n-1}(a_{2n}^2 - a_1^2)$

(C) $\frac{n}{n+1}(a_1^2 - a_{2n}^2)$ (D) None of these

Q9. Consider an A.P. with first term a and common difference is not zero. Let S_n denotes the sum of its n-terms. If $\frac{S_{kx}}{S_x}$ is idenpendent of x, then S_n=

(A) n^2a (B) n^3a

(C) $2n^2a$ (D) $(n^2+n)a$

Properties of A.P.

Q10. If $a^2(b+c), b^2(c+a), c^2(a+b)$ are in AP, then a, b, c are in :-

(A) AP (B) GP

(C) HP (D) None of these

Q11. If $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in AP, then a^2, b^2, c^2 will be in-

(A) AP (B) GP

(C) HP (D) None of these

Q12. If $<a_n>$ is an A.P. and $a_1 + a_4 + a_7 +$ .....+ $a_{16} = 147$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to:-

(A) 96 (B) 98

(C) 100 (D) None

Arithmetic Mean

Q13. If x, y, z are in A.P. and A.M. of x and y is a and that to y and z is b, then A.M. of a and b is :-

(A) x (B) y

(C) z (D) $\frac{1}{2}(x+y)$

Q14. The sum of 40 A.M's between two number is 120. The sum of 50 A.M.'s between them is equal to:-

(A) 130 (B) 160

(C) 150 (D) None of these

Q15. If $A_1, A_2, A_3, \ldots\ldots, A_{45}$ are arithmetic means inserted between the numbers a and b then the value of $\left(\frac{b+A_{45}}{b-A_{45}}\right)-\left(\frac{A_1+a}{A_1-a}\right)$ is :-

(A) 102 (B) 88 (C) 92 (D) 90

Geometric Progression

Q16. The eight term of G.P. is 128 and common ratio is 2. The product of its first five terms is :-

(A) 4^6 (B) 4^3 (C) 4^5 (D) 4^4

Q17. The sum of first three terms of a G.P. is to the sum of next three terms is 125 : 27. The common ratio of G.P. is :-

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$ (C) $\frac{5}{3}$ (D) None

Q18. $8 + 88 + 888 + \ldots.$ to n terms –

(A) $\frac{80}{81}(10^n-1)-\frac{8n}{9}$ (B) $\frac{10}{81}(10^n-1)$

(C) $\frac{80}{81}(10^n-1)+\frac{8n}{9}$ (D) None of these

Q19. The sum of an infinite G.P. is 23 and the sum of squares of the series is 69, then first term is :-

(A) $\frac{69}{13}$ (B) $\frac{10}{3}$ (C) $\frac{9}{10}$ (D) $\frac{13}{69}$

Q20. In an infinite G.P., the sum of first three terms is 70. If first & third terms are multiplied by 4 and second term is multiplied by 5, the resulting terms form an A.P., then the sum of infinite G.P. is :-

(A) 120 (B) 40 (C) 160 (D) 80

Q21. If $1 + r + r^2 + \ldots\ldots + r^n = (1 + r)(1 + r^2)(1 + r^4)(1 + r^8)$, then the value of n is :-

(A) 13 (B) 14

(C) 15 (D) 16

Q22. If $r > 1$ and $x = a + \frac{a}{r} + \frac{a}{r^2} + \ldots..\text{to } \infty$,

$y = b - \frac{b}{r} + \frac{b}{r^2} - \ldots\text{to } \infty$ and

$z = c + \frac{c}{r^2} + \frac{c}{r^4} + \ldots\text{to } \infty$, then $\frac{xy}{z} =$

(A) $\frac{ab}{c}$ (B) $\frac{ac}{b}$

(C) $\frac{bc}{a}$ (D) None of these

Q23. The sum of 40 terms of the series $1 + 2 + 3 + 4 + 5 + 8 + 7 + 16 + 9 + \ldots..$ is :-

(A) $398 + 2^{20}$ (B) $398 + 2^{21}$

(C) $398 + 2^{19}$ (D) none of these

Q24. If G be the GM between x and y, then the value of $\frac{1}{G^2-x^2}+\frac{1}{G^2-y^2}$ is equal to :-

(A) G^2 (B) $\frac{2}{G^2}$ (C) $\frac{1}{G^2}$ (D) $3G^2$

Q25. If product of 'n' geometric means between $\frac{1}{2}$ and 32 is 1024 then n = ?

(A) 10 (B) 5 (C) 3 (D) 8

Harmonic Progression

Q26. The pth term T_p of H.P. is $q(p + q)$ and q^{th} term T_q is $p(p + q)$ when $p > 2, q > 2$. Then which of the following is true.

(A) $T_{p+q} = pq$ (B) $T_{pq} = p + q$

(C) $T_{p+q} > T_{pq}$ (D) All of these

AM, GM, HM

Q27. If p, q be two AM's and G be one GM between two numbers, then G^2 is equal to

(A) $(2p-q)(p-2q)$

(B) $(2p-q)(2q-p)$

(C) $(2p-q)(p+2q)$

(D) None of these

Q28. If x be the AM and y, z be two GM's between two positive numbers, then $\frac{y^3+z^3}{xyz}$ is equal to :-

(A) 1 (B) 2 (C) 3 (D) 4

Q29. If A.M. between p and q ($p \geq q$) is two times the G.M. then p : q is -

(A) 1 : 1

(B) 2 : 1

(C) $(2+\sqrt{3}):(2-\sqrt{3})$

(D) 3 : 1

Q30. If sum of A.M. and H.M between two positive numbers is 25 and their GM is 12, then sum of numbers is :

(A) 9 (B) 18

(C) 32 (D) 18 or 32

Q31. For all positive values of x and y, the value of $\frac{\left(1+x+x^2\right)\left(1+y+y^2\right)}{xy}$ is :-

(A) < 9 (B) ≤ 9 (C) > 9 (D) ≥ 9

Q32. If $x^2 + y^2 + z^2 = 1$, where $x, y, z \in R$, then maximum value of $x^2.y^3.z^4$ is :-

(A) $\frac{2^5}{3^{\frac{15}{2}}}$ (B) $\frac{2^{10}}{3^{15}}$ (C) $\frac{2^{10}}{3^{\frac{9}{2}}}$ (D) $\frac{2^{15}}{3^9}$

AGP

Q33. Let $S=\frac{4}{19}+\frac{44}{19^2}+\frac{444}{19^3}+\ldots$ upto ∞ then S is equal to :-

(A) $\frac{38}{81}$ (B) $\frac{4}{19}$ (C) $\frac{36}{171}$ (D) $\frac{44}{79}$

Q34. If $|x| < 1$ then $1 + 3x + 5x^2 + 7x^3 + \ldots$. upto ∞ is equal to :-

(A) $\frac{1+x}{(1-x)^2}$ (B) $\frac{2+x}{(1-x)^2}$

(C) $\frac{x}{(1-x)^2}$ (D) None of these

Special Sequence - 1

Q35. If the sum of the first n natural numbers is 1/5 times the sum of the their squares, then the value of n is -

(A) 5 (B) 6 (C) 7 (D) 8

Special Sequence - 2

Q36. 99th term of the series

2 + 7 + 14 + 23 + 34 + ... is-

(A) 9998 (B) 9999

(C) 10000 (D) 100000

Q37. Sum of n terms of series 12 + 16 + 24 + 40 +..... will be

(A) $2(2^n - 1) + 8n$

(B) $2(2^n - 1) + 6n$

(C) $3(2^n - 1) + 8n$

(D) $4(2^n - 1) + 8n$

Q38. If the sum of $1+\frac{1+2}{2}+\frac{1+2+3}{3}$..... to n terms is S, then S is equal to :-

(A) $\frac{n(n+3)}{4}$ (B) $\frac{n(n+2)}{4}$

(C) $\frac{n(n+1)(n+2)}{6}$ (D) n^2

Special Sequence - 3

Q39. The sum to n terms of the series

$\frac{3}{1^2}+\frac{5}{1^2+2^2}+\frac{7}{1^2+2^2+3^2}+\ldots\ldots$ is

(A) $\frac{3n}{n+1}$ (B) $\frac{6n}{n+1}$

(C) $\frac{9n}{n+1}$ (D) $\frac{12n}{n+1}$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If sum of n terms of a sequence is given by $S_n = 3n^2 - 5n + 7$ & t_r represents its rth term, then -

(A) $t_7 = 34$ (B) $t_2 = 7$

(C) $t_{10} = 34$ (D) $t_8 = 40$

Q2. Let $a_1, a_2, a_3, \ldots\ldots$ and $b_1, b_2, b_3, \ldots\ldots$ be arithmetic progression such that $a_1 = 25$, $b_1 = 75$ and $a_{100} + b_{100} = 100$, then -

(A) The common difference in progression 'a_i' is equal but opposite in sign to the common difference in progression 'b_j'.

(B) $a_n + b_n = 100$ for any n.

(C) $(a_1 + b_1), (a_2 + b_2), (a_3 + b_3), \ldots\ldots$ are in A.P.

(D) $\sum_{r=1}^{100} (a_r + b_r) = 10^4$

Q3. $0.4 + 0.44 + 0.444 + \ldots\ldots$ to 2n terms -

(A) $\frac{4}{81}[18n - 1 + 100^{-n}]$

(B) $\frac{4}{81}[18n + 1 + 100^{-n}]$

(C) $\frac{4}{18}[18n - 1 + 100^{-n}]$

(D) $\frac{4}{81}[18n - 1 + 10^{-2n}]$

Q4. Indicate the correct alternative(s), for $0 < \phi < \frac{\pi}{2}$, if $x = \sum_{n=0}^{\infty} \cos^{2n}\phi$, $y = \sum_{n=0}^{\infty} \sin^{2n}\phi$ and $z = \sum_{n=0}^{\infty} \cos^{2n}\phi \sin^{2n}\phi$, then -

(A) $xyz = xz + y$ (B) $xyz = xy + z$

(C) $xyz = x + y + z$ (D) $xyz = yz + x$

Q5. a, b, c are three distinct real numbers, which are in G.P. and $a + b + c = xb$, then -

(A) $x < -1$ (B) $-1 < x < 2$

(C) $2 < x < 3$ (D) $x > 3$

Q6. If a, b, c are non-zero numbers such that $3(a^2 + b^2 + c^2 + 1) = 2(a + b + c + ab + bc + ca)$, then a, b, c are

(A) in A.P. (B) in G.P.

(C) in H.P. (D) all equal

Q7. If a, b, c are in H.P. then -

(A) $\frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c}$ are in H.P.

(B) $\frac{2}{b} = \frac{1}{b-a} + \frac{1}{b-c}$

(C) $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ are in G.P.

(D) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P.

Q8. If 10 harmonic means $H_1, H_2, H_3 \ldots\ldots H_{10}$ are inserted between 7 and $-\frac{1}{3}$, then -

(A) $H_1 = -7$ (B) $H_2 = \frac{3}{7}$

(C) $H_1 = -\frac{1}{7}$ (D) $H_{10} = -\frac{7}{19}$

Q9. Let a, x, b be in A.P. ; a, y, b be in G.P. and a, z, b be in H.P. If $x = y + 2$ and $a = 5z$ then-

(A) $y^2 = xz$ (B) $x > y > z$

(C) $a = 9, b = 1$ (D) $a = \frac{9}{4}, b = \frac{1}{4}$

Q10. For the series

$$S = 1 + \frac{1}{(1+3)}(1+2)^2 + \frac{1}{(1+3+5)}(1+2+3)^2 + \frac{1}{(1+3+5+7)}(1+2+3+4)^2 + \ldots\ldots$$

(A) 9th term is 25
(B) 7th term is 16
(C) sum of first ten terms is 505/4
(D) sum of first ten terms is 505

Q11. If t_n be the nth term of the series $1 + 3 + 7 + 15 + \ldots\ldots$, then -

(A) $t_5 + 1 = 32$
(B) $t_7 = 2^7 + 1$
(C) $t_{10} = 2^{10} - 1$
(D) $t_{100} = 2^{50} + 1$

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. Four options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q12. Let $< a_n >$ be an increasing arithmetic progression with $a_1 = 2$ and $7.a_{a_2} = 4.a_{a_5}$, then match List-I with List-II and select the correct answer using the code given below the list.

List-I	**List-II**
(P) $5(a_2 - a_1)$ is equal to	(1) 75
(Q) $\sum_{r=1}^{10} a_r$ is equal to	(2) 10
(R) $\sum_{r=1}^{10} (a_{r+1} - a_r)$ is equal to	(3) 5
(S) $\sum_{r=1}^{10} a_{a_r}$ is equal to	(4) 65

Codes :	P	Q	R	S
(A)	4	3	2	1
(B)	3	4	2	1
(C)	3	4	1	2
(D)	4	3	1	2

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13.

Column - I	**Column - II**
(A) 10th term of sequence $S = 1 + 5 + 13 + 29 + \ldots\ldots$is	(P) 1620
(B) The sum of all two digit numbers which are not divisible by 2 or 3 is	(Q) 1/3
(C) The sum of $\frac{5}{1^2.4^2} + \frac{11}{4^2.7^2} + \frac{17}{7^2.10^2} + \ldots\ldots$is	(R) 2045
(D) If first two terms of H.P. is $\frac{1}{2}$ & $\frac{1}{3}$ then the harmonic mean of first four terms is	(S) $\frac{240}{77}$

TRIPLE MATCH COLUMN TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Q14.

	Column - I Series	**Column - II** T_n (n^{th} Term)	**Column - II** S_n(Sum of n terms)
(1)	$1.3.5 + 3.5.7 + 5.7.9 + ...$	(A) $T_n = \frac{1}{2}(3n^2 - n + 4)$	(P) $S_n = 3.2^n - 2n - 3$
(2)	$3 + 7 + 14 + 24 + ...$	(B) $T_n = 3\left(\frac{2^n}{2}\right) - 2$	(Q) $S_n = n(2n^3 + 8n^2 + 7n - 2)$
(3)	$1 + 4 + 10 + 22 + ...$	(C) $T_n = \frac{n^2}{4} + \frac{n}{2} + \frac{1}{4}$	(R) $S_n = \frac{n}{2}(n^2 + n + 4)$
(4)	$\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + ...$	(D) $T_n = (2n - 1)(2n + 1)(2n + 3)$	(S) $S_n = \frac{n}{4}\left[(2+n) + \frac{(n+1)(2n+1)}{6}\right]$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q15. If (666.....(1000) times)^2 + (888.....(1000) times) = (444.....K times) then K = ?

Q16. Let a_1, a_2, a_3 be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer 'n' for which $a_n < 0$ is :-

Q17. $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{6} + ...100$ terms $= 2^{-P} + q$, then $|p - q| = ?$

ASSERTION & REASON TYPE

Q18. Suppose four distinct positive numbers a_1, a_2, a_3, a_4 are in G.P. Let $b_1 = a_1$, $b_2 = b_1 + a_2$, $b_3 = b_2 + a_3$ and $b_4 = b_3 + a_4$.

Statement-I : The numbers b_1, b_2, b_3, b_4 are neither in A.P. nor in G.P.

Statement-II : The numbers b_1, b_2, b_3, b_4 are in H.P.

(A) Statement-I is true, Statement-II is true; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true; Statement-II is NOT a correct explanation for Statement-I.

(C) Statement-I is true, Statement-II is false.

(D) Statement-I is false, Statement-II is true.

Q19. **Statement-I :** If a, b, c are three positive numbers in G.P., then

$$\left(\frac{a+b+c}{3}\right)\cdot\left(\frac{3abc}{ab+bc+ca}\right)=\left(\sqrt[3]{abc}\right)^2.$$

Statement-II : (A.M.) (H.M.) = $(G.M.)^2$ is true for any set of positive numbers.

(A) Statement-I is true, Statement-II is true; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true; Statement-II is NOT a correct explanation for Statement-I.

(C) Statement-I is true, Statement-II is false.

(D) Statement-I is false, Statement-II is true.

SUBJECTIVE TYPE

Q20. The sum of n terms of two arithmetic series are in the ratio of $(7n + 1) : (4n + 27)$. Find the ratio of their n^{th} term.

Q21. In an AP of which 'a' is the Ist term, if the sum of the Ist p terms is equal to zero, show that the sum of the next q terms is $-\left(\frac{aq(p+q)}{p-1}\right)$.

Q22. If $S_1, S_2, S_3, \ldots S_n, \ldots$ are the sums of infinite geometric series whose first terms are $1, 2, 3, \ldots n, \ldots$ and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n+1}, \ldots$ respectively, then find the value of $\sum_{r=1}^{2n-1} S_r^2$.

Q23. If $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ form first three terms of an A.P., then find the sum $1 + \sin 2x + \sin^2 2x + \ldots\infty$.

Q24. If $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is HM of a and b, where $a \neq b$ then $2n^2 = ?$

Q25. If one AM 'a' and two GM's p and q be inserted between any two given numbers then show that $p^3 + q^3 = 2apq$.

Q26. The value of $x + y + z$ is 15, if a, x, y, z, b are in A.P. while the value of; $(1/x) + (1/y) + (1/z)$ is 5/3 if a, x, y, z, b are in H.P. Find a & b.

Q27. If number of coins earned in n^{th} game is $n2^{n+2} - 2^n$ and total number of coins earned in first 10 games is $10(B.2^{10} + 1)$, where $B \in N$, then the value of B is :

Q28. Find the sum of infinite terms of the sequence :

$$1+2\left(1+\frac{1}{n}\right)+3\left(1+\frac{1}{n}\right)^2+4\left(1+\frac{1}{n}\right)^3+\ldots\ldots$$

Q29. Find the sum of the n terms of the sequence

$$\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\ldots\ldots$$

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
C	C	D	B	B	B	A	C	B	A
11	12	13	14	15	16	17	18	19	20
C	A	B	C	C	C	B	D	B	A
21	22	23	24	25	26	27	28	29	30
C	B	A	A	D	B	A	A	A	C
31	32	33	34	35	36	37	38	39	40
C	A	A	A	C	C	A	D	A	B
41									
B									

Ex - 1A

1. (B,C) **2.** (B,D) **3.** (A) **4.** (A,C) **5.** (B,C,D)

6. (A,C) **7.** (B) **8.** (B) **9.** (A) **10.** C

11. C **12.** B **13.** A **14.** D

15. (A $\rightarrow$ R ; B $\rightarrow$ Q, ; C $\rightarrow$ P)

16. (A) $\rightarrow$ (Q), (B) $\rightarrow$ (P), (C) $\rightarrow$ (P)

17. 0 **18.** (a) Not Required (b) Not Required

19. 3 **20.** Not Required **21.** 2 **22.** 33470

23. Not Required **24.** $\sqrt{\frac{3}{2}(n^2+3n)}$

Ex - 2

1	2	3	4	5	6	7	8	9	10
B	C	A	A	B	C	C	A	A	A
11	12	13	14	15	16	17	18	19	20
A	B	B	C	D	C	B	A	A	D
21	22	23	24	25	26	27	28	29	30
C	A	B	C	B	D	B	B	C	C
31	32	33	34	35	36	37	38	39	
D	A	A	A	C	A	D	A	B	

Ex - 2A

1. (A,D) **2.** A, B, C, D **3.** (A,D) **4.** (B,C) **5.** A, D

6. (A,B,C,D) **7.** (A,B,C,D) **8.** (A,D) **9.** (A,B,C) **10.** (A,B,C)

11. A, C **12.** (B)

13. (A $\rightarrow$ R ; B $\rightarrow$ P ; C $\rightarrow$ Q ; D $\rightarrow$ S)

14. (1) $\rightarrow$ D $\rightarrow$ Q; (2)$\rightarrow$ A $\rightarrow$ R; (3) $\rightarrow$ B $\rightarrow$ P; (4) $\rightarrow$ C $\rightarrow$ S

15. 2000 **16.** 25 **17.** 1 **18.** C

19. C **20.** $(14n - 6)/(8n + 23)$ **21.** Not required

22. $\frac{n(2n+1)(4n+1)}{3} - 1$ **23.** 2 **24.** 2

25. Not required **26.** a = 1 b = 9 or a = 9 b = 1

27. 7 **28.** n^2 **29.** $\frac{n(n+1)}{2(n^2+n+1)}$

-: Important Notes & Formulas :-